CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of M. Sc. (Biochemistry) Examination, May, 2019

Course code & Name BC 761 & Bioenergetics and Metabolism

Date: 01/05/2019 Day: Wednesday Time: 1.30 PM to 2.00 PM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	Inborn errors of metabolism are referred to as		01
	(i) congenital metabolic diseases	(ii) inherited metabolic diseases	
	(iii) both A and B	(iv) none of above	
2.	The creatine kinase reaction is:		01
	(i) irreversible	(ii) slow compared with glycolysis	
	(iii) inhibited by low pH in the muscle	(iv) not activated until all the ATP has been used up	
3.	All are conditions of ketosis except one		01
	(i) Starvation	(ii) Uncontrolled diabetes mellitus	
	(iii) Von Gierke’s disease	(iv) High carbohydrate diet	
4.	Which of the following is a coenzyme in the reaction catalyzed by glyceraldehyde 3-phosphate dehydrogenase?		01
	(i) NAD^+	(ii) Cu^+	
	(iii) Heme	(iv) ATP	
5.	An enzyme used in both Glycolysis and Gluconeogenesis is:		01
	(i) Glucose 6-phosphatase	(ii) 3-phosphoglycerate kinase	
	(iii) Hexokinase	(iv) Phosphofructokinase-1	

6.	The end product of fatty acid synthesis in mammals is		01
	(i) Arachidonic acid	(ii) Palmitic acid	
	(iii) Linoleic acid	(iv) Stearic acid	
7.	The oxidation of 3 mol of glucose by the pentose phosphate pathway may result in the production of:		01
	(i) 3 mol of Pentose, 6 mol of NADPH, and 3 mol of CO_2	(ii) 2 mol of Pentose, 4 mol of NADPH, and 8 mol of CO_2	
	(iii) 3 mol of Pentose, 4 mol of NADPH, and 8 mol of CO_2	(iv) 4 mol of Pentose, 3 mol of NADPH, and 3 mol of CO_2	
8.	The first few seconds of the 800-meter race will likely feel good to Himani, but she will find that the length of the event will tax her ____________.		01
	(i) glycolytic system	(ii) 1mol ATP/ 1mol PCr	
	(iii) the presence of ADP	(iv) more carbohydrates	
9.	Muscle lactate production increases when:		01
	(i) oxygen is readily available	(ii) pyruvate cannot be formed from glucose breakdown	
	(iii) the pH of the muscle falls	(iv) glycolysis is activated at the onset of exercise	
10.	Which of the following is a common compound shared by the TCA cycle and the Urea cycle?		01
	(i) Fumarate	(ii) α-keto glutarate	
	(iii) Oxaloacetate	(iv) Sucinyl CoA	
11.	Name the most active organs in the animal body which have the ability to synthesize triacylglycerol?		01
	(i) Spleen	(ii) Kidney	
	(iii) Adipose tissue	(iv) Liver and intestines	
12.	This system also uses the molecule phosphocreatine to rebuild ATP and maintain a relatively constant supply. The __________ enzyme acts on PCr to separate Pi from creatine. The energy released can then be used to couple Pi to an ADP molecule, forming ATP.		01
	(i) the presence of ADP	(ii) glycolytic system	
	(iii) 1mol ATP/ 1mol PCr	(iv) creatine kinase	
13.	Which of the following is an autoimmune disease?		01
	(i) Sickle cell anemia	(ii) Type 1 Diabetes mellitus	
	(iii) Haemophilia A	(iv) Type 2 Diabetes mellitus	
14.	The following are all characteristics of metabolic syndrome:		01
	(i) Hypertension, hypoglycemia, and high HDL	(ii) Intra-abdominal obesity, hypertension, and low HDL	
	(iii) Dyslipidemia, intra-abdominal obesity, and insulin insensitivity	(iv) Intra-abdominal obesity, hypotension, and low LDLs	
15.	Glucose enters muscle cells mostly by:		01
	(i) Facilitated diffusion using a specific glucose transporter.	(ii) Co-transport with amino acids	
	(iii) Simple diffusion	(iv) Co-transport with sodium	

16.	Which enzyme often mal functions in diseases associated with the symptoms of high blood triglyceride levels and Steatorrhea?		01
	(i) Phospholipase D	(ii) Thiokinase	
	(iii) Lipoprotein lipase	(iv) Pancreatic lipase	
17.	Which of the following is not an intermediate of the citric acid cycle?		01
	(i) Acetoacetate	(ii) Oxalosuccinate	
	(iii) Citrate	(iv) Succinyl-CoA	
18.	A36-year-old male has been diagnosed with type 1 hyperlipidemia. Which of the following lipoprotein concentration is elevated in such disorder?		01
	(i) VLDL	(ii) LDL	
	(iii) IDL	(iv) Chylomicrons	
19.	What reactions need “activation energy” to get started?		01
	(i) Coupled reactions	(ii) Exergonic reactions	
	(iii) Endergonic reactions	(iv) Endergonic & exergonic reactions	
20.	Biologically, which of the following has the highest redox potential?		01
	(i) Oxygen	(ii) Proteins	
	(iii) Lipids	(iv) Carbohydrates	

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II Semester of M. Sc. (Biochemistry) Examination, May, 2019

Course code & Name BC 761 & Bioenergetics and Metabolism

Date: 01/05/2019 Day: Wednesday Time: 2.01 PM to 4.30 PM Maximum Marks: 50

Instructions:
1. **Section I and II must be attempted in TWO ANSWER SHEETS.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of a non - programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q – II Answer the following questions as directed (Any 5)		**20**
1.	What are Inborn Errors of Metabolism? What are the different types of inborn errors of metabolism? Discuss briefly.	**1+3**
2.	Justify the reasoning that glutamic acid plays a pivotal role in the metabolism of amino acids.	**04**
3.	Why is it important to have different pathways for glycogenesis and glycogenolysis in liver and muscle cells? Justify your answer.	**04**
4.	For the hydrolysis of 1 mole of ATP to ADP at 37 °C, the standard free enthalpy change $\Delta \boldsymbol{G}^{\boldsymbol{0}} = -35$ kJ . mol^{-1}. Calculate the free enthalpy change $\Delta \boldsymbol{G}$ at the ratio ATP/ADP = 100:1. (Temperature 37^0C, R = 8.3143 J K^{-1} mol $^{-1}$. Concentrations of water and inorganic phosphate are to be omitted from the equilibrium equation, assuming that they do not change significantly)	**04**
5.	Discuss the clinical significance of transaminases.	**04**
6.	What are autoimmune liver diseases? Explain briefly any one of them.	**04**
7.	What are the symptoms and treatments of Hyperammonemia?	**2+2**
Q-III Answer the following questions as directed (Any 10)		**30**
1.	The Pentose phosphate pathway is active in liver, adipose tissue,mammary gland. Why this the pathway is less active in non-lactating mammary gland and skeletal muscle?	**03**
3.	What is a metabolic disease? How do you recognize a metabolic disorder ?	**03**

4.	What is Cori cycle? How is it differ from Glucose Alanine cycle?	**1+2**
5.	What is the source of $NH4^+$ produced during amino acid catabolism? Name the three ketone bodies.	**2+1**
6.	Explain why a galactosemic child maintained on a galactose free diet manifests normal growth and development?	**03**
7.	A pregnant woman who has extremely lactose intolerant asked her physician if she could not drink milk or dairy products. What advice should be given?	**03**
8.	Consider the reaction, $A \Longleftrightarrow B + B$, where ΔG^o is zero (a) Explain, in general, how entropy may change during the catabolic reaction depicted above. (b) Determine what the sign of the free energy change will be if the concentrations of all the species are raised above the standard conditions by 2-fold. Circle your answer from the three listed, and show your assumptions and equations used to justify your response (A) NEGATIVE (B) N SIGN / ZERO (C) POSITIVE	**1+2**
9.	Write a short note on Tyrosinemia.	**03**
10.	H^+ moves from the intermembrane space, through ATP synthase, and into the mitochondrial matrix. Which term best describes this process: facilitated diffusion or active transport?	**03**
11.	What is the pH inside the matrix of the mitochondrion relative to the intermembrane space? Explain in your answer how the pH is affected during respiration.	**1+2**
12.	In mammals, glucose is the only fuel that the brain uses under non starvation conditions and the only fuel that red blood cells can use at all. There are many carbohydrates, why is glucose instead of some other monosaccharides such a prominent fuel?	**2+1**
13.	We know that gluconeogenesis is not simply a reversal of glycolysis. Why is this so? In your explanation be sure to mention at least two of the important steps and the enzymes involved.	**03**
14.	How does the body metabolize alcohol? Discuss briefly.	**03**